

### 1. Understanding the concepts

*Mark the correct statements by ■□ and the incorrect ones by □■. You learn from your mistakes.*

- The empty set  $\emptyset$  can be made into a vector space  $(\emptyset, \oplus, \odot)$  by an appropriate choice of  $\oplus$  and  $\odot$ .
- The maps  $\oplus$  and  $\odot$  of a vector space  $(V, \oplus, \odot)$  map two elements of  $V$  to one element of  $V$ .
- The set  $V$  underlying a vector space  $(V, \oplus, \odot)$  must contain at least one element.
- If both  $(V, \oplus, \odot)$  and  $(V, \boxplus, \boxdot)$  are vector spaces, then they are inevitably identical.
- If both  $(V, \oplus, \odot)$  and  $(W, \oplus, \odot)$  are vector spaces, then inevitably  $V = W$ .
- If  $(V, \oplus, \odot)$  is a vector space, a bi-linear map  $\mu : V \times V \rightarrow V$  satisfies  $\mu(\lambda \odot v, \lambda \odot v') = \lambda \odot \mu(v, v')$ .
- Every vector space  $(V, \oplus, \odot)$  carries an inner product defined by  $\odot$ .
- A  $(p, q)$ -tensor  $T$  is a multi-linear map  $T : \underbrace{V \times \dots \times V}_p \times \underbrace{V^* \times \dots \times V^*}_q \rightarrow V$ .

### 2. Mastering the formalism

*Mastership is in the basics. Work hard to find the required short proofs. This exercise will pay off.*

- (a) Prove that a vector space  $(V, \oplus, \odot)$  has only one neutral element  $O$ .

Solution: Assume that  $O$  and  $O'$  are two neutral elements. Then

$$O \stackrel{N^\oplus}{=} O \oplus O' \stackrel{C^\oplus}{=} O' \oplus O \stackrel{N^\oplus}{=} O',$$

*which shows that they are already equal. Thus there is only one neutral element.*

- (b) Prove that for every linear map  $\xi : V \rightarrow W$  between vector spaces  $(V, \oplus, \odot)$  and  $(W, \boxplus, \boxdot)$

$$\xi(O_V) = O_W,$$

where  $O_V$  denotes the neutral element in  $V$  and  $O_W$  the neutral element in  $W$ .

Solution: Due to linearity of  $\xi$  one has

$$\xi(O_V) \stackrel{N^\oplus}{=} \xi(O_V \oplus O_V) = \xi(O_V) \boxplus \xi(O_V),$$

*from which one already concludes with  $N^\boxplus$  that  $\xi(O_V) = O_W$ .*

- (c) Consider three vector spaces  $(V, \oplus, \odot)$ ,  $(W, \boxplus, \boxdot)$  and  $(U, \boxplus, \boxdot)$  and two linear maps

$$\varphi : V \rightarrow W \quad \text{and} \quad \psi : W \rightarrow U.$$

Prove that their composition, defined by

$$\psi \circ \varphi : V \rightarrow U, \quad (\psi \circ \varphi)(v) := \psi(\varphi(v))$$

for any  $v \in V$ , is again a linear map.

Solution: For any  $v, v' \in V$  and  $\lambda \in \mathbb{R}$ , it follows from the linearity of  $\psi$  and  $\varphi$  that

$$(\psi \circ \varphi)(v \oplus v') = \psi(\varphi(v \oplus v')) = \psi(\varphi(v) \boxplus \varphi(v')) = \psi(\phi(v)) \boxplus \psi(\varphi(v')) = (\psi \circ \varphi)(v) \boxplus (\psi \circ \varphi)(v'),$$

which shows additivity of  $\psi \circ \varphi$ , while

$$(\psi \circ \varphi)(\lambda \odot v) = \psi(\varphi(\lambda \odot v)) = \psi(\lambda \boxtimes \varphi(v)) = \lambda \diamond \psi(\varphi(v)) = \lambda \diamond (\psi \circ \varphi)(v)$$

shows the scaling property. Together this amounts to the claimed linearity of  $\psi \circ \varphi$ .

(d) Prove, from our definition of an inner product  $g$  on a vector space  $(V, \oplus, \odot)$ , that

$$g(v, v) = 0 \quad \text{if and only if} \quad v = O$$

and that the inequality

$$|g(v, v')|^2 \leq g(v, v)g(v', v')$$

holds for all  $v, v' \in V$ .

Solution: From the definition of an inner product, we know that  $g(v, v) > 0$  for all  $v \neq O$ . It thus suffices to observe that from the bilinearity of  $g$  one has

$$g(O, O) \stackrel{N^\oplus}{=} g(O \oplus O, O) = g(O, O) + g(O, O),$$

from which one immediately concludes that  $g(O, O) = 0 \in \mathbb{R}$ , which proves the first assertion. In order to prove the inequality, which is known as the Cauchy-Schwarz inequality, we start from the only known equality

$$0 \leq g(v \oplus \lambda \odot v', v \oplus \lambda \odot v') = g(v, v) + 2\lambda g(v, v') + \lambda^2 g(v', v')$$

an observe that choosing

$$\lambda = -\frac{g(v, v')}{g(v', v')}$$

yields

$$0 \leq g(v, v) - \frac{g(v, v')^2}{g(v', v')},$$

whose rearrangement yields the claimed inequality. Do not worry if you did not find this proof. If you did, you are really good at this already.

(e) Prove that for any linear map  $\varphi : V \rightarrow W$  between vector spaces  $(V, \oplus, \odot)$  and  $(W, \boxplus, \boxtimes)$ , the set

$$\ker \varphi := \{v \in V \mid \varphi(v) = O_W\} \quad (\text{the so-called kernel of the map})$$

is a vector subspace of  $V$ , which is the case by definition if for all  $a, b \in \ker \varphi$  and  $\lambda \in \mathbb{R}$  also

$$\lambda \odot a \in \ker \varphi \quad \text{and} \quad a \oplus b \in \ker \varphi.$$

Solution: Assume  $a, b \in \ker \varphi$  and  $\lambda \in \mathbb{R}$ , so that  $\varphi(a) = \varphi(b) = 0$ . But then also

$$\varphi(a \oplus b) = \varphi(a) \boxplus \varphi(b) = O_W \quad \text{and} \quad \varphi(\lambda \odot a) = \lambda \boxtimes \varphi(a) = O_W,$$

which shows that  $a \oplus b$  and  $\lambda \odot a$  lie in  $\ker \varphi$ .

(f) Let  $(V, \oplus, \odot)$  be a vector space and  $\varphi : V \rightarrow V^*$  and  $\psi : V \rightarrow V$  be linear maps. Prove that

$$\pi : V \times V \rightarrow \mathbb{R}, \quad \pi(v, v') := \varphi(v)(\psi(v'))$$

is a  $(0, 2)$ -tensor.

Solution: One needs to show linearity in each slot. Denoting the vector space operations on  $V^*$  by  $\boxplus$  and  $\boxminus$ , one has, due to linearity of  $\varphi$  and  $\psi$ , additivity in the first slot,

$$\pi(v \oplus w, v') = \varphi(v \oplus w)(\psi(v')) = (\varphi(v) \boxplus \varphi(w))(\psi(v')) = \pi(v, v') + \pi(w, v')$$

for all  $v, v', w \in V$ , and additivity in the second slot,

$$\pi(v, v' \oplus w') = \varphi(v)(\psi(v' \oplus w')) = \varphi(v)(\psi(v') \oplus \psi(w')) = \pi(v, v') + \pi(v, w')$$

for all  $v, v', w' \in V$ . Similarly one has the scaling property in the first slot,

$$\pi(\lambda \odot v, v') = \varphi(\lambda \odot v)(\psi(v')) = (\lambda \boxminus \varphi(v))(\psi(v')) = \lambda \pi(v, v'),$$

and the scaling property in the second slot,

$$\pi(v, \lambda \odot v') = \varphi(v)(\psi(\lambda \odot v')) = \varphi(v)(\lambda \odot \psi(v')) = \lambda \pi(v, v').$$

for all  $v, v' \in V$  and  $\lambda \in \mathbb{R}$ . Hence  $\pi : V \times V \rightarrow \mathbb{R}$  is multilinear and thus a  $(0, 2)$ -tensor.